

GOULBURN, Australia

WMO: 957160

Lat: 34.8086S

Lon: 149.7311E

Elev: 2102

StdP: 13.61

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	26.4	18.8	15.3	40.6	22.2	18.0	39.0	32.5	46.0	30.3	47.0	1.0	0	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.4	92.9	63.7	88.6	62.7	84.5	62.0	68.4	80.4	66.9	78.2	65.5	76.4	15.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	101.2	71.0	63.6	95.1	69.8	62.1	90.0	68.6	33.9	80.5	32.6	78.4	31.5	76.4	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.4	25.9	22.9	DB	18.6	98.7	2.5	3.9	16.8	101.5	15.3	103.8	13.9	106.0	12.0	108.8	(4)
(5)			WB	17.8	71.3	2.6	3.6	15.9	73.9	14.4	76.0	12.9	78.0	11.0	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	68.1	66.8	62.0	55.0	48.2	44.0	42.5	44.2	49.1	54.2	59.8	64.7	(6)	
	DBStd	10.43	6.41	5.88	5.51	5.37	4.89	4.33	4.09	4.58	4.95	5.81	6.75	6.32	(7)	
	HDD50	821	0	0	1	16	93	184	236	189	74	23	5	0	(8)	
	HDD65	4084	42	44	125	301	520	629	699	645	476	336	183	84	(9)	
	CDD50	2577	560	469	371	167	38	5	2	9	48	154	298	456	(10)	
	CDD65	364	137	93	31	1	0	0	0	0	0	2	26	74	(11)	
	CDH74	5071	1770	1002	425	62	0	0	0	0	7	133	553	1119	(12)	
	CDH80	2014	840	400	121	10	0	0	0	0	1	20	194	428	(13)	
(14) Wind	WSAvg	9.4	9.8	9.3	8.6	7.3	8.3	8.6	9.8	10.6	10.6	10.0	10.1	10.1	(14)	
(15) Precipitation	PrecAvg	24.7	2.4	2.3	2.2	2.0	1.8	1.8	1.8	2.2	2.0	2.3	2.6	2.2	(15)	
	PrecMax	35.9	7.1	6.6	7.1	8.2	4.9	7.3	3.8	8.5	3.9	5.8	4.6	5.2	(16)	
	PrecMin	14.3	0.1	0.1	0.1	0.0	0.1	0.4	0.2	0.2	0.2	0.2	0.2	0.0	(17)	
	PrecStd	5.8	1.6	1.8	1.6	2.1	1.4	1.5	1.0	1.9	0.9	1.3	1.3	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.8	95.0	89.1	81.7	69.3	61.8	60.4	67.2	73.9	82.9	92.1	95.1	(19)	
		MCWB	64.6	65.2	61.2	59.4	54.0	51.3	50.5	52.2	54.5	58.1	60.8	62.0	(20)	
	2%	DB	94.2	89.9	83.2	74.7	65.5	57.5	56.6	61.7	69.5	77.7	85.3	89.4	(21)	
		MCWB	64.4	63.8	61.1	57.2	54.1	50.1	47.9	49.4	53.5	57.3	60.5	61.8	(22)	
	5%	DB	89.4	84.8	78.7	71.0	62.2	55.0	53.8	58.1	65.7	73.4	80.3	84.9	(23)	
		MCWB	63.5	63.3	60.5	56.2	52.5	49.0	46.8	48.0	52.3	56.1	59.8	60.8	(24)	
	10%	DB	84.2	80.2	74.4	67.3	59.0	52.9	51.3	54.9	61.9	69.0	75.5	80.5	(25)	
		MCWB	63.4	63.1	60.0	55.2	51.5	48.0	45.4	46.9	51.0	55.0	58.1	60.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.5	70.7	67.9	62.6	59.1	54.7	53.5	54.6	58.3	62.7	66.3	68.0	(27)	
		MCDB	85.2	83.6	74.8	72.6	62.7	58.2	57.0	62.4	67.1	74.3	78.4	79.4	(28)	
	2%	WB	68.5	68.4	65.4	60.2	56.5	52.1	49.8	51.4	56.0	60.2	63.7	65.8	(29)	
		MCDB	82.8	78.8	74.1	69.3	61.5	55.2	53.8	58.6	64.9	70.9	76.8	77.4	(30)	
	5%	WB	67.0	66.7	63.6	58.5	54.3	50.3	48.0	49.4	54.0	58.2	62.0	64.1	(31)	
		MCDB	80.2	76.3	72.1	65.8	59.9	53.7	52.0	56.0	62.4	68.8	74.3	76.7	(32)	
	10%	WB	65.6	65.2	62.1	56.9	52.4	48.6	46.4	47.4	52.2	56.5	60.2	62.6	(33)	
		MCDB	77.7	74.7	69.9	64.2	57.8	51.9	50.4	53.7	59.8	66.3	71.1	75.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	25.4	22.5	22.2	23.7	22.1	18.2	18.6	21.1	22.9	24.7	24.7	25.5	(35)	
		MCDBR	37.1	32.2	31.2	29.9	27.8	22.7	23.6	27.3	31.2	33.7	35.3	34.5	(36)	
	5% WB	MCWBR	11.4	10.4	12.1	14.3	16.7	15.6	15.8	16.3	16.9	15.6	13.9	11.3	(37)	
		MCWBR	27.1	24.1	22.4	23.2	21.5	17.5	18.2	23.1	25.0	26.9	27.7	26.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.364	0.352	0.320	0.298	0.284	0.274	0.268	0.277	0.305	0.323	0.346	0.340	(40)	
		taud	2.438	2.503	2.575	2.614	2.603	2.641	2.657	2.617	2.520	2.489	2.459	2.502	(41)	
	Ebn at Noon		308	305	304	294	282	277	286	298	304	311	312	317	(42)	
		Edh at Noon	38	35	30	25	23	20	21	25	31	35	37	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		2197	1875	1586	1247	952	725	833	1118	1538	1873	2034	2242	(44)	
	RadStd		167	168	139	108	52	58	60	74	115	161	174	161	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	-173	+277	+113
Regional (35 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	-2	-149	+313	+148

Nomenclature: See separate page